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Bio

Bioinformatician and Biotechnology M.Sc. graduate working at the intersection of clinical genomics, cancer functional genomics, and computational biology. Currently serving as a Bioinformatician at Ecegen Genetic Diseases Assessment Center, supporting genomic data analysis, variant interpretation, and reproducible NGS workflows. Completed master's research at Bezmialem Vakif University, Cingoz Lab, investigating metabolic mechanisms associated with TRAIL resistance in glioblastoma through pooled CRISPR/Cas9 screening, sequencing readout analysis, MAGECK-based gene prioritization, pathway analysis, and integration of transcriptomic and metabolomic data. Experienced in Linux-based bioinformatics workflows, Python and R data analysis, clinical variant annotation, cancer genomics, functional genomics, molecular modeling, and experimental molecular biology.

Education

- M.Sc. Bezmialem Vakif University**, Biotechnology (100% Scholarship) Sept 2024 – July 2026
- **GPA:** 3.92/4.00
 - **Master's Thesis:** Bioinformatic Analysis of a CRISPR/Cas9-Based Metabolic Gene Screen to Investigate TRAIL Resistance in Glioblastoma
 - **Research Focus:** Functional genomics, pooled CRISPR/Cas9 screening, MAGECK-based gene prioritization, glioblastoma metabolism, therapeutic resistance, transcriptomics, metabolomics, and pathway analysis
 - **Research Support:** TÜBİTAK 3501 Career Development Support Program
- B.Sc. Kadir Has University**, Bioinformatics and Genetics (100% English) Sept 2017 – June 2023
- **GPA:** 2.68/4.00
 - **Project:** Computational and in Vitro Analysis of GAT-3 Structure and Function as a Potential Therapeutic Target
 - **Coursework:** Bioinformatics, Computational Drug Design, Molecular Modeling & Simulations
- B.Sc. Universidad de A Coruña**, Biology Sept 2020 – June 2021
- **Program:** Erasmus Program
 - **Project:** Computational Analysis of Viral Mutations
 - **Relevant Coursework:** Molecular Techniques, Genomic Analysis Techniques, Statistics
 - Completed coursework in an international academic environment with instruction in Spanish.

Experience

- Ecegen Genetic Diseases Assessment Center**, Bioinformatician Jan 2026 – present
- Support genomic data analysis and interpretation workflows for clinical and research-oriented genetic testing, including processing and evaluation of FASTQ, BAM/CRAM, BED, and VCF data.
 - Perform and support NGS workflows involving quality control, alignment, variant processing, annotation, filtering, prioritization, and evidence assessment.
 - Work with bioinformatics tools and genomic resources including GATK, SAMtools, bcftools, IGV, Ensembl VEP, SnpEff, ANNOVAR, ClinVar, dbNSFP, and related annotation databases.

- Develop and improve reproducible Linux-based workflows for genomic analysis, quality control, variant interpretation, clinical data handling, and report generation.
- Contribute to bioinformatics pipeline development, workflow automation, structured genomic data handling, and integration between clinical genetics and computational analysis systems.

Bezmialem Vakif University Life Science and Biotechnology Institute, Cingoz Lab, Project Assistant / Graduate Researcher Sept 2024 – June 2026

- **TÜBİTAK 3501:** Conducted research within the Scientific and Technological Research Council of Türkiye Career Development Support Program.
- Investigated metabolic mechanisms associated with TRAIL resistance in glioblastoma using functional genomics, cell-based experiments, metabolomics, transcriptomics, and computational analysis.
- Analyzed a pooled CRISPR/Cas9 metabolic screen covering 2,981 genes with 10 sgRNAs per gene, including sequencing readout processing, quality assessment, sgRNA-level analysis, and candidate gene prioritization.
- Applied MAGeCK RRA to identify depleted and enriched genes across sensitive, resistant, and baseline experimental comparisons.
- Performed functional and pathway-level interpretation using GO, KEGG, Reactome, and gene-set enrichment approaches.
- Integrated CRISPR screening results with metabolomics, transcriptomics, and TCGA-GBM data to investigate biological mechanisms associated with therapy resistance.
- Investigated metabolic processes including urea cycle and arginine biosynthesis, lipid and carboxylic acid metabolism, ion transport, and redox regulation.
- Performed mammalian cell culture, cell viability assays, nucleic-acid handling, agarose gel electrophoresis, and metabolomics sample preparation.

Istinye University Genetic Diseases Diagnostic Center, Intern June 2022 – Sept 2022

- Gained hands-on experience in clinical genetics and cytogenetic diagnostic workflows under the supervision of laboratory specialists.
- Reviewed genomic datasets in VCF, BAM, and BED formats and performed genomic region navigation and annotation assessment.
- Conducted mutation detection and microsatellite instability (MSI) analysis using SeqScape in cancer genomics applications.

Biofarma Pharmaceutical Company, Intern June 2020 – Aug 2020

- Reviewed and interpreted scientific literature related to pharmaceutical research, drug development, and therapeutic discovery.
- Collaborated with research and regulatory teams and gained exposure to pharmaceutical development, testing, and compliance workflows.

Technical Skills

NGS & Clinical Genomics: FASTQ, BAM, CRAM, BED, and VCF processing; FastQC, fastp, Trimmomatic, BWA, HISAT2, SAMtools, bcftools, GATK, IGV, SnpEff, ANNOVAR, Ensembl VEP, ClinVar, and dbNSFP. Experienced in variant calling, annotation, filtering, prioritization, quality control, and genomic interpretation

Functional Genomics & Cancer Bioinformatics: Pooled CRISPR/Cas9 screening, MAGeCK RRA, sgRNA-level analysis, gene prioritization, TCGA analysis, GO, KEGG, Reactome, gene-set enrichment, pathway analysis, glioblastoma biology, therapy resistance, and tumor metabolism

Transcriptomics & Multi-omics: RNA-seq differential expression analysis, lncRNA analysis, DESeq2, fgsea, clusterProfiler, msigdb, transcriptomic interpretation, metabolomics integration, and multi-omics analysis

Programming & Data Analysis: Python, R, Bash, SQL, Pandas, NumPy, Biopython, Matplotlib, ggplot2, Bioconductor, statistical analysis, data wrangling, automation, and scientific visualization

Linux & Bioinformatics Workflows: Linux/Ubuntu, Bash, Docker, Nextflow, nf-core, Conda/Mamba, Git, GitHub, environment management, command-line data processing, reproducible computational workflows, and large-scale genomic data handling

Computational Drug Design: Structure-based computational therapeutic discovery, protein and ligand preparation, molecular docking, virtual screening, binding pose inspection, and ligand-receptor interaction analysis using AutoDock and AutoDock Vina

Molecular Modeling & Simulations: Molecular dynamics simulations using NAMD and structural visualization and analysis using PyMOL and VMD. Experienced in protein stability assessment, conformational analysis, structural comparison, superposition, and protein-ligand interaction analysis

Laboratory Skills

- **Cell Culture:** maintenance, passaging, cryopreservation, and experimental handling of mammalian cell lines
- **Cell Viability Assays:** MTT-based experimental workflows
- **Basic Molecular Biology:** DNA/RNA isolation, quantification, and quality assessment
- **Agarose Gel Electrophoresis:** DNA/RNA fragment analysis and visualization
- **Metabolomics Sample Preparation:** cell extraction workflows for LC-MS/GC-MS-based metabolomics
- **Animal-Model Research Experience:** mice and rats; certification can be provided

Research Interests

Clinical Bioinformatics • Cancer Genomics • Functional Genomics • CRISPR Screening • Glioblastoma • Therapy Resistance • Tumor Metabolism • Precision Oncology • Variant Interpretation • Translational Bioinformatics • Reproducible Genomic Workflows

Languages

English: Proficient (IELTS Academic: 6.0/9.0)

Spanish: Limited Working Proficiency

Turkish: Native

References

Available upon request.